

Isaac Nathan da Silva Barbosa

Full-Stack AI Engineer · Security Researcher · Full-Stack · Python · React · Cloud · Threat Intel
Build intelligent systems that connect AI, cloud, and product — from LLM-orchestrated pipelines to production deploy.

PWD — ASD (ICD-11: 6A02.2) + ADHD (ICD-11: 6A05.2)

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TECHNICAL SKILLS

AI / ML: GPT-4.1 · Azure OpenAI · PyTorch · DeepForest · OpenCV · Ollama · DALL-E 3 · Flux · Sora · Veo 3.1 · Gemini SDK · scikit-learn

Backend: Python · FastAPI · Node.js · Express · Java · Azure Cosmos DB · PocketBase · SQL

Frontend: React 19 · TypeScript · JavaScript · Flutter · Tailwind CSS v4 · Vite · React Native (Expo 54)

Cloud & Infra: Azure · Docker · Git / GitHub · PocketBase · Linux (CachyOS/Hyprland) · PyInstaller / Inno Setup

Integrations: Mercado Pago PIX · Canva Connect API · Placid API · Templated.io · Pexels API · NVIDIA NIM API

Offensive Security: OSINT · Google Dorking · Shodan · Certificate Transparency · Passive DNS · Nmap · Wireshark · Burp Suite · SNMP Enum · WordPress Audit · Directory Listing · Auth Bypass · Base91 Decoder · JS Deobfuscation

Defensive Security: LGPD · WAF Verification · TLP Protocol · Responsible Disclosure (CERT.br/CTIR Gov) · CVE Analysis · Post-Fix Verification · NIST CSF · OWASP Top 10 · MITRE ATT&CK; (awareness)

Threat Intelligence: IOC Extraction · Base91/JS Deobfuscation · IOC Sharing (AbuseIPDB, URLhaus) · Phishing Infra Takedown · Pipeline (Browser → JS → Decoder → IOC → CERT.br)

Other: Fluent English · Agile Methodologies · API Testing · FastAPI + Jinja2 · Rich CLI · xlwings/COM

PROFESSIONAL SUMMARY

From Industrial Chemistry to Computer Engineering: first Python in 2022 (Thonny IDE, no GPT, extra afternoon classes). Built ForestAI from scratch, manually annotating drone images from Fundação Renova. Freelance contract with Meritage Homes (USA) — migrated 500GB to Azure Cosmos DB with zero downtime, architected AI pipelines (GPT-4.1, DALL-E 3, Flux) and cross-platform chatbot infra. At Paware, architected LLM-orchestrated pipelines for HelloSocial (GPT-4.1, DALL-E 3, Flux). I learn by solving real problems — from cross-platform MIME types to ReAct agents with schema enforcement.

PROFESSIONAL EXPERIENCE

Security Researcher & Threat Intelligence | Mar 2026 — Present

Security Researcher (Independent / Responsible Disclosure) | Remote

- 20+ vulnerabilities in Brazilian government/sector infrastructure — 5 fixes confirmed via CERT.br/CTIR Gov (Mar–Jun 2026). Three disclosure waves: zero retention, multi-channel notification (CERT.br + CTIR Gov + direct), post-fix verification.
- Microsoft/OneDrive phishing takedown: toppled origin server (procorereviews.com) via controlled ~10k-request flood (sustained HTTP 521). Rotated infra (kochcnfontainer.vu) mapped, IOCs extracted, reported to CERT.br/Cloudflare.
- Reversed 3 JS files obfuscated with custom Base91 (3 unique alphabets, 176 strings). Pipeline: Browser → JS download → Custom Base91 decoder (Python) → IOC extraction → CERT.br/Cloudflare report. (Full detail in Security Disclosures entry below.)
- Methodology NIST CSF + LGPD: Passive Recon (Dorking, Shodan, CT logs) → Exposure Verification → Documentation → Secure Deletion (shred) → Multi-channel Notification → Post-fix Follow-up.

Full-Stack Developer with AI focus | Oct 2025 — May 2026

Paware Softwares | Remote

- Migrated hundreds of GB of legacy datasets from Google Drive to Azure Cosmos DB — automated extraction, compression, injection into panels for WhatsApp agent. Docker + schema validation with auto-rollback.
- Resolved cross-platform MIME type issue (Android vs iOS) — validation layer that became the backbone of the Cosmos DB migration, with extensive testing on iOS devices.
- Architected image-generation pipeline for HelloSocial — multi-step pipeline with GPT-4.1 (generation, curation, composition, validation), Flux Kontext Pro + DALL-E 3 + Placid/Canva, local SQLite/PIL fallback.

Software Engineer (Freelance) | Nov 2025 — Dec 2025

Meritage Homes (USA) | Remote

- Migrated 500GB of critical data to Azure Cosmos DB with zero downtime — multi-region architecture, eventual consistency, auto-failover.
- Architected AI pipelines (GPT-4.1, DALL-E 3, Flux) for creative asset generation — copywriting, image, video — integrated into the builder's marketing workflow.
- Built cross-platform chatbot infrastructure (iOS/Android) with context-aware intent routing, PT/ES/EN translation.
- Engineered PDF parsing pipelines for LLM ingestion with Spanish/Portuguese translation/transcription.
- Conducted E2E probing and UAT simulating realtors' workflows — real-time feedback to stakeholders.

Robotics Instructor | Sep 2025 — Oct 2025

SuperNerds | Mariana, MG

- Learned Arduino and LEGO in under 2 weeks and taught robotics to children/teens.
- Balanced mornings here with evening work at Paware — discipline and adaptability.

EDUCATION

Computer Engineering | 2024 — 2029

Cruzeiro do Sul

5th semester — in progress

Chemical Engineering | 2022 — 2024

UFSJ — Campus Alto Paraopeba

Transition — self-directed AI study

PROJECTS

Security Disclosures & Threat Intel · github.com/xAngryBadger/security-disclosures

OSINT · Nmap · Burp Suite · Shodan · Base91 Decoder · CERT.br/CTIR Gov · LGPD · NIST CSF

- 20+ vulnerabilities in Brazilian government/sector infrastructure — 5 fixes confirmed via CERT.br/CTIR Gov (Mar–Jun 2026). Three disclosure waves: zero retention, multi-channel notification, post-fix verification.
- Microsoft/OneDrive phishing takedown: origin server toppled via sustained ~10k requests (HTTP 521). Rotated infra mapped, IOCs extracted, reported to CERT.br/Cloudflare.
- Threat Intel Pipeline: Browser → JS download → Custom Base91 decoder (3 alphabets, 176 strings) → IOC extraction (PageConfig, tokens, OneDrive redirect) → CERT.br report.
- Methodology NIST CSF + LGPD Art. 46/48: Passive Recon → Exposure Verification → Documentation → Secure Deletion (shred) → Multi-channel Notification → Post-Fix Verification → Documented Closure.

HarplA · github.com/xAngryBadger/harplA

Python · GPT-4.1 · DALL-E 3 · Flux 2.0 Pro · Sora · Veo 3.1 · Azure Cosmos DB · SQLite

- Creative automation engine with 7+ AI models — multi-model pipeline with GPT-4.1: copywriting, image search, design compositing, video generation. Lightweight stack by default (SQLite + local PIL) with fallback to paid APIs.
- 6,900+ LOC async Python with security tests and zero hardcoded secrets. Backend swap: local SQLite to Azure Cosmos DB + Blob Storage, toggled via env var. Cron-ready with file-locking and stuck batch recovery.

Urutau · github.com/xAngryBadger/flora-sensus

Flutter · Dart · Drift/SQLite · React · PocketBase · TypeScript

- Flutter offline-first app for forest inventory with custom sync service — cascading UUID remapping across related tables and transactional rollback on failure.
- Architecture and sync logic built from scratch; code generated with LLM assistance (web) and manually reviewed. React admin panel with auth, photos, reports, and XLSX/PDF/CSV export.

ForestAI · github.com/xAngryBadger/forestai

PyTorch · DeepForest · OpenCV · scikit-learn · TensorBoard

- Forest species detection and classification with Deep Learning — built from scratch without AI-assisted coding. Stack Overflow + Thonny IDE only.
- Manual annotation of drone images from Fundação Renova (bounding boxes), DeepForest training on local GPU, stratified splits. TensorBoard curve interpretation — detecting memorization vs generalization.

Fennec Excel · github.com/xAngryBadger/Sahara-Fenneck

Python · Ollama · CustomTkinter · xlwings/COM · PyInstaller · Inno Setup

- Local AI assistant for Excel via ReAct agent (Ollama/qwen2.5). Natural language commands to filter, sort, rename sheets and manipulate spreadsheets with auto-checkpoint before every change.
- 6+ OAuth integrations (Gmail, Teams, Calendar, Drive, Outlook, Trello) with user confirmation before modifications. Native Windows installer (Inno Setup + PyInstaller). Bilingual PT/EN interface.

Orca · github.com/xAngryBadger/orca

Python · pandas · FastAPI · Jinja2 · Rich CLI · openpyxl · unittest

- Operational planning engine for large-scale forest restoration — automatic generation of executive dossiers with crew allocation, territory mapping, and schedules.
- Tariff and operational cost management. FastAPI + Jinja2 + Rich CLI interface with unit test suite.

MaineCoon · github.com/xAngryBadger/minneapolis

Node.js · mineflayer · NVIDIA NIM API · Reinforcement Learning

- Minecraft bot with natural language commands via LLM (NVIDIA NIM API) — mine, craft, follow, navigate, and interact via chat. Reinforcement learning module for autonomous behavior.

HelloSocial

Python · FastAPI · Azure OpenAI · Flux · Canva Connect · React

- AI-powered social media post creation and scheduling platform — project at Paware that inspired HarplA. Image generation pipeline with Flux Kontext Pro and DALL-E 3, copy and template agents.

Capivara · github.com/xAngryBadger/capivara

React 19 · TypeScript · FastAPI · pypdf · PyMuPDF · cloudflare

- Full PDF suite with 15 tools — DOCX/XLSX→PDF, PDF→DOCX, compress, merge, split, rotate, watermark, page numbers, header/footer, protect, unlock, OCR, PDF→images, PDF/A. React 19 frontend, FastAPI backend with StreamingResponse.
- Free serverless tunneling via cloudflare + Google Colab — zero account, zero token. Lazy imports for heavy deps (PyMuPDF, pytesseract, pikepdf). pypdf 4+ with PdfWriter-based merge.

Tarsier · github.com/xAngryBadger/tarsier

React 19 · TypeScript · Vite · Tailwind CSS v4 · Framer Motion

- 100% client-side JSON workbook with collapsible tree, transformations (pretty-print, minify, CSV, TypeScript types) and stats. Zero backend — everything runs in the browser.

Kakapo · github.com/xAngryBadger/kakapo

React 19 · Canvas API · TypeScript · Vite · Tailwind CSS v4

- Browser image editor — filters, crop, resize, rotation, text overlay, adjustments. Commit-based editing engine with undo/redo. 100% client-side, native Canvas API.

Oilbird · github.com/xAngryBadger/oilbird

React 19 · TypeScript · FastAPI · WeasyPrint · cloudflare

- Markdown → PDF converter with live split-pane preview and professional WeasyPrint rendering. FastAPI + WeasyPrint backend with CSS Paged Media (@page, headers/footers), free cloudflare tunnel via Google Colab.

CERTIFICATIONS

Python Essentials 1 & 2 — Cisco Networking Academy — *Fundamentals + advanced Python (OOP, libraries, cert prep).* **IT Security** — Bradesco Foundation — *Corporate infrastructure and data protection.*

Data Science Essentials with Python — Cisco Networking Academy — *Pandas, Matplotlib — hands-on project-based learning.*
Data Analytics Essentials — Cisco Networking Academy — *Industry-recognized analytics tools.*
AI for Process Optimization — Escola Virtual Gov · Enap · Serpro (71h) — *Gov AI Nucleus (PBIA) — strategic AI in public management and infosec.*
Networking Basics (120h) — Cisco Networking Academy — *Completed during Computer Engineering.*
Intro to Cybersecurity — Cisco Networking Academy — *Network security and cyber threats.*

Fluent English (3 years) — KUMON — *Technical documentation and meetings with international teams.*
eJPTv2 / eCPPTv2 / OSCP — INE / eLearnSecurity — *In progress / Planned 2026 — Hands-on pentesting, AD, pivoting, evasion.*
eWPTX / eWPTXv2 — eLearnSecurity — *Planned 2026 — Advanced web app pentesting (WAF bypass, auth bypass, business logic).*
Cloud Security (AZ-500 / CCSP) — Microsoft / (ISC)² — *Planned 2026 — Azure security, identity, data, governance.*
MITRE ATT&CK; Defender (MAD) — MITRE — *Planned 2026 — ATT&CK; mapping, detection engineering, threat-informed defense.*

LANGUAGES

Portuguese (native) · English (fluent, KUMON 3 years)

Full case studies, demos & source code -> xangrybadger.github.io/isaac-vitae